DRIVEABILITY—COMPRESSED NATURAL GAS (CNG)—ROUGH IDLE OR NO START WHEN COLD—VEHICLES EQUIPPED WITH CNG BI-FUEL ENGINE ONLY

Article No. 01-23-1

FORD: 1996-2000 CONTOUR
1998-1999 E SERIES
1998-2001 F-150

ISSUE

Due to possible regional Fuel Quality issues in some areas, it may be necessary to clean the low flow injectors in the Compuvalve, and the associated fuel introduction hardware between the Compuvalve and intake manifold (fuel rails, injector sleeves). Typical symptoms of contamination in the low flow injectors are:

- Rough run in alt fuel mode most noticeable at idle, runs ok in gasoline mode
- No start (green light on) when vehicle is cold in alt fuel mode, but will start if vehicle has been previously warmed up in gasoline mode
- No start in alt fuel mode (with green light on) at any time, but starts in gasoline mode.

ACTION

Verify Condition. If normal diagnostics cannot pinpoint the condition, clean the Low Flow CNG injectors in the compuvalve. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE
THIS SERVICE PROCEDURE SHOULD ONLY BE PERFORMED BY AN AUTHORIZED CNG FUEL SYSTEM TECHNICIAN.

NOTE
FOLLOW ALL CNG FUEL SYSTEM SAFETY PROCEDURES LISTED IN THE APPROPRIATE WORKSHOP MANUAL.

1. Set fuel selector switch to Alt position.
2. If the vehicle cannot be started, in alt fuel mode mode, the low flow injectors may be completely stuck closed. If this is the case, attempt to free them by applying 12 Volts directly to the injector coil:
   a. Disconnect the low flow injector harness (J14).
   b. Apply battery voltage to pins A and D (marked on the side of the connector) to open injector #1. Listen for an audible click to confirm that the injector has opened.

NOTE
THE INJECTORS ARE NOT DESIGNED TO HAVE A CONTINUOUS 12 V SUPPLY (NORMAL OPERATION IS ABOUT 1.5 V). PROLONGED EXPOSURE TO 12 V WILL CAUSE PERMANENT DAMAGE, SO LIMIT THE 12 VOLTS TO A MAXIMUM OF 5 SECONDS.

   c. If no click is heard, try several more times, always for 5 seconds or less (wait at least 5 seconds between application of voltage).
   d. Repeat the above step for injector #2 applying voltage to pins B and D. If vehicle can be started continue to Step 3.
   e. If the vehicle can still not be started after this procedure, replace the Compuvalve.
   f. Follow service procedure listed in appropriate Workshop Manual for Compuvalve replacement.
   g. Leak check only the connections removed for Compuvalve replacement.
   h. Restart engine, verify the problem has been corrected. Return vehicle.
3. Start and run engine to a minimum temperature of 75°C (165°F) (MST or ECT depending on vehicle model year) to ensure engine stability.
4. Shut off AC and Defrost.
5. Disconnect the electrical connection to the CNG tank valve and run the engine until it auto switches to gasoline.
6. Following procedure in GFI Service Manual, bleed off residual CNG line pressure.

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7. Remove coalescent filter body and drain any contaminants. Wipe the inside with a clean cloth.
8. Fill to within 1/4 inch of the top with isopropyl alcohol with coalescent filter in place (alcohol must be at least 90% by volume).
9. Re-install coalescent filter body (ensure filter element is still in place).
   - Torque for Filter Body: 40 ± 6 N•m (30 ± 5 lb-ft)
   - Torque for Drain Plug: 10 ± 3 N•m (8 ± 2 lb-ft)

CAUTION
ENSURE THAT ANY LINES THAT WERE DISCONNECTED IN STEP "F" ARE RE-CONNECTED.

10. Re-connect the CNG tank valve harness.
11. Key on, check for leaks at the coalescent filter housing.
12. Start and run vehicle.
13. Hook up NGS and monitor RPM.
14. Increase the RPM to between 1150 and 1300, maintain this RPM.
15. Note time, run for 15 minutes.
16. After 15 minutes, shut off engine.
17. Restart engine, verify the problem has been corrected, if it has not been resolved, repeat the procedure.

LABOR OPERATION CLAIMING CHART			
Operation	**Labor Description**	**Vehicle**	**Time**
973809A	Apply voltage to low flow injectors then perform fuel system cleaning procedure.	E-Series	1.5
		F-150/F-250 LD	1.4
		Contour	1.4
973809B	Apply voltage to low flow injectors then replace compuvalve (fuel system cleaning procedure not required).	E-Series	1.6
		F-150/F-250 LD	1.6
		Contour	1.2

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

DEALER CODING

BASIC PART NO.	CONDITION CODE
9F593	42

OASIS CODES: 602300, 603300, 607000, 607400, 607500, 608000, 608400, 608500, 609000, 609400, 609500, 610000, 610500, 611000, 611500, 612000, 612500, 614000, 614500, 698298